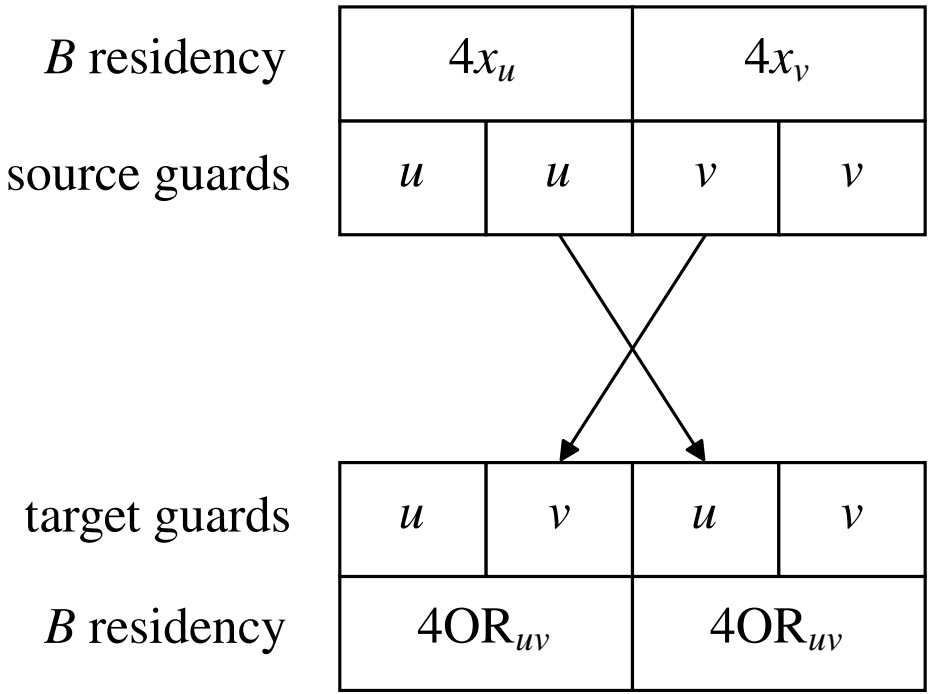

One A word and one B word live at a time.

Only the middle event identities interchange.

$$\Delta B = 8\mathrm{OR}_{uv} - 4x_u - 4x_v = 4(x_u \oplus x_v)$$